



Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN
2026

Bachelor in Computer Applications
Course Title: OOP in Java
Code No: CACS 204
Semester: III

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions.

[6×5 = 30]

2. What is conditional operator? Write a program to find smallest number among three number using command line argument. [1+4]
3. What is a constructor? Explain its types with examples. [1+4]
4. Explain different types of Access controls used in Java. [5]
5. What is a multithreaded program? How multithreaded programs can be created in java? Explain with example. [1+4]
6. Explain the use of static keyword in Java with proper example. [5]
7. Define exception handling. Differentiate between the use of throw and throws with proper example. [1+4]
8. Write short notes on (Any two): [2.5+2.5]
 - a. Abstract class
 - b. Applet life cycle
 - c. JDBC

Group C

Attempt any TWO questions.

[2×10 = 20]

9. What is an interface? What is the use of it? Explain the use of interface to implement multiple inheritance in Java with proper example. [1+3+6]
10.
 - a. What is inheritance? Explain its types. [1+4]
 - b. Define character stream. Write a program to copy the content of a file to another file using character streams. [1+4]
 - a. Create a GUI form in Java that three numbers input from the user and displays the smallest number in a label when Check button is pressed. [5]
 - b. Differentiate between AWT and swing. Explain any six swing controls with their uses. [2+3]



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Bachelor in Computer Applications
Course Title: Data Structure & Algorithm
Code No: CACS 201
Semester: III

Full Marks: 60
Pass Marks: 24
Time: 3 hours

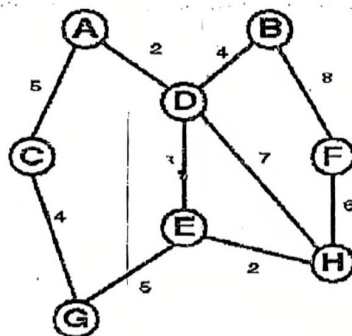
Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions.

[6×5 = 30]

2. What is Deterministic and Non-deterministic Algorithm? How are data structure classified? [2.5+2.5]
3. Write an algorithm for the insertion and deletion operations performed on the circular queue.
4. Explain different types of linked list with example. •
5. Write down recursive algorithm to find Fibonacci series. •
6. Explain the different traversal techniques used in binary tree with examples. •
7. Find MST for following graph using Kruskal algorithm.



8. Explain different collision resolution techniques with examples. •

Group C

Attempt any TWO questions.

[2×10 = 20]

9. Why stack is called a LIFO data structure? Write an algorithm to convert infix expression to postfix expression and show how it works with your own example • [1+9]
10. What is AVL tree? Construct AVL tree from given data set: 3,5,11,8,4,1,12,7,2,6 [5+5]
11. How quick sort implements divide and conquer strategy. Sort the following numbers using heap sort : 46, 25, 35, 49, 10, 92, 32



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Bachelor in Computer Applications

Course Title: Web Technology

Code No: CACS205

Semester: III

Full Marks: 60

Pass Marks: 24

Time: 3 hours

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions.

[6×5 = 30]

2. Explain different types of lists in HTML including their related elements and attributes. [5]
3. What is element? Explain the structure of a webpage using semantic elements. [1+4]
4. What is multi-tier architecture? What are challenges can arise when implementing multi-tier systems? [1+4]
5. What is DTD? Explain different types of DTD. [2+3]
6. What is XSL? Explain any four XML elements used in XSLT. [1+4]
7. Develop a web page that accept user CV from user using file control and upload into "files" folder using server-side language. [2+3]
8. Which HTTP error belong to web client? Differentiate between web server and client. [1+4]

Group C

Attempt any TWO questions.

[2×10 = 20]

9. Write HTML code to create a form for student registration for an **entrance form fill-up system**. The form should include fields for: full name, email, phone number, address, date of birth, gender, and course applied for. Also, write server-side code to validate the form with required field validation for all fields, type validation for phone number, and store the submitted information into a **"registrations"** table within an **"entrance"** database. [10]
10. Design an XML document to store **"flight ticket booking information"**, including fields such as Booking ID, passenger name (first name, middle name, last name), flight number, departure city, destination city, travel date, ticket class, and total baggage. Also, write XML Schema (XSD) code to validate the structure of this data with the following rules: [4+4+2]
 - a. Flight number must follow the pattern: one uppercase alphabet followed by three digits (e.g., A123).
 - b. Total baggage must be restricted to a value between 15 kg and 25 kg.
11. a. Explain different types of CSS with proper example. [4]
b. Write HTML code to display following internal marks evaluation structure using table with dummy data. [6]

SN	TU Reg No.	Student Name	Internal Marks Evaluations			
			Presentation (5)	Attendance (5)	MidTerm Exam(5)	Pre-Board Exam(5)
1						
2						



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Bachelor in Computer Applications
Course Title: System Analysis and Design
Code No: CACS 203
Semester: III

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions. [6×5 = 30]

2. What do you mean by system? Compare the various sources of software. [2.5+2.5]
3. What is a project charter? How does it differ from a Baseline Project Plan? [2.5+2.5]
4. Explain the process to transform ER diagram into relational table with suitable example.
5. Define feasibility study? Explain the types of feasibility. [1+4]
6. Describe several methods for interacting with the system.
7. Explain the different approaches of installation?
8. Write short notes on [2.5+2.5]
 - a) Decision table and decision tree
 - b) CASE tool

Group C

Attempt any TWO questions. [2×10 = 20]

9. What is meant by "balancing" in DFDs? Draw the DFD of Library Management System up to 2nd level. [2+8]
10. What are the skills required for a project manager? Explain different phases of project management? Explain the types of maintenance with suitable example. [3+3+4]
11. Describe the project identification and selection process. Differentiate between black box and white box testing. [5+5]



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Bachelor in Computer Applications
Course Title: Probability and Statistics
Code No: CAST 202
Semester: III
Candidates are required to answer the questions in their own words as far as possible.

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Group B

Attempt any SIX questions.

[6×5 = 30]

2. What do you understand by data? Distinguish between primary and secondary data.
3. The following table represents the marks of 100 students

Marks	0 - 20	20 - 40	40 - 60	60 - 80	80 - 100
Number of students	14	-	27	-	15

If the mean is 48, find the missing frequencies.

4. Define correlation. In order to find the correlation coefficient between two variable X and Y from 13 pairs of observations, following calculation are made:

$$\sum X = 30, \sum Y = 5, \sum X^2 = 670, \sum Y^2 = 285, \sum XY = 334$$

On subsequent verification, it was found that the pair (11, 4) was copied wrongly. The correct being (10, 14), find the correct value of correlation coefficient.

5. Define regression. Estimate the blood pressure using the given data when age is 65.

Age(in years)	56	42	72	36	63	47
Blood pressure	147	125	160	118	149	128

6. Define Poisson distribution and state the conditions under which it is applicable. A telephone switch board of a company receives an average of 4 phone calls in 10 minutes interval. What is the probability of receiving (i) exactly 3 calls in a 10 minutes interval (ii) at most 2 calls in half an hour interval.
7. Suppose the sample standard deviation of response times (in milliseconds) for a web server is 12 ms. If we want to estimate the average response time with 99% confidence and margin of error of 3 ms, how many observations should be included in the sample?
8. Define sampling. Distinguish between census survey and sample survey.

Group C

Attempt any TWO questions.

[2×10 = 20]

9. What are the various methods of measuring dispersion in Statistics? In two factories A and B engaged in similar type of industry, the average weekly wages and variance are given below:

	Factory A	Factory B
Average weekly wages (Rs)	186	175
Variance of wages	81	100
No. of wage earners	500	600

- i. Which factory pays larger amount as weekly wages?
- ii. Which factory shows greater variability in the distribution of weekly wages?
- iii. What is the mean, standard deviation and variance of all the workers' wages in two factories taken together?
10. The number of emails sent per week by employees of a software company follows a normal distribution with mean 80 and standard deviation 16. Find: (i) the probability that an employee sends at least 100 emails in a week (ii) the probability that an employee sends at most 120 emails in a week. (iii) the probability that an employee sends between 85 and 95 emails in a week.
11. The following table gives the data on the performance of three different detergents at three different water temperatures. The performance was obtained on the 'whiteness' reading based on specially designed equipment for nine loads of washing:

	Detergent A	Detergent B	Detergent C
Cold water	45	43	55
Warm water	37	40	56
Hot water	42	44	46

Perform a two-way analysis of variance, using the level of significance 0.05.